

LUNCHENG ZHANG

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Summary

Graduate Mechanical Engineering student at Oregon State University specializing in advanced manufacturing and electric vehicle systems. Experienced in EV performance simulation, CNC machining, and robotic system modeling. Proficient in MATLAB, Python, and CAD tools, with strong skills in engineering analysis and process optimization. Seeking entry-level roles in EV or manufacturing engineering.

Education

Oregon State University

Master of Engineering in Mechanical Engineering (Advanced Manufacturing)

Expected: Aug 2026

Xinlian College, Henan Normal University, China

Bachelor of Civil Engineering

Graduate: Aug 2022

Technical Skills

CAD & Design: SolidWorks (3D modeling, assembly), AutoCAD

Manufacturing: CNC machining, G-code programming, NC

programming GD&T: Interpretation of engineering drawings and tolerances

Welding: Basic welding techniques and weld joint types

Materials: Material selection and engineering analysis

EV Systems: Electric powertrain fundamentals, battery basics, drive cycle analysis

Programming: Python, MATLAB

Process Improvement: Lean manufacturing, VSM, 5 Whys

Tools: NumPy, Excel

Projects

Electric Vehicle Performance Simulation (MATLAB, EV Systems)

- * Simulated EV performance using standard driving cycles (US06, SC03)
- * Calculated motor torque, power, and efficiency under varying conditions
- * Evaluated energy consumption and regenerative braking behavior

Robotic Arm Modeling and Kinematics Analysis (Python, NumPy)

- * Designed a robotic arm system with defined joint constraints
- * Applied transformation matrices to simulate motion and positioning
- * Modeled system behavior under varying input conditions
- * Enhanced understanding of kinematic behavior and system response through simulation

CNC Machining and G-code Programming (Manufacturing)

- * Developed and modified G-code for machining operations
- * Optimized feed rate and spindle speed for improved performance
- * Evaluated machining accuracy and surface quality
- * Improved machining efficiency and consistency through parameter adjustment

Lean Manufacturing and Process Optimization (VSM, 5 Whys)

- * Analyzed production processes using lean manufacturing principles
- * Identified inefficiencies and proposed workflow improvements
- * Applied VSM and root cause analysis methods
- * Reduced non-value-added steps and improved process clarity

Experience

Student Trainer – Arnold Dining Center, Oregon State University

- * Trained 10+ new employees on operational procedures and safety standards
- * Supervised daily workflow in a high-volume environment
- * Coordinated team tasks to improve operational efficiency
- * Improved team coordination and reduced service delays

Field Engineer – China Construction Fifth Engineering Division

- * Assisted with construction operations and on-site coordination
- * Ensured compliance with engineering standards and safety requirements

Architectural Designer – Zhongfan International Design Co.

- * Created AutoCAD drawings for architectural projects
- * Reviewed structural feasibility and design accuracy